



Alfred Sherwood Romer
A **palaeontologist** par excellence, comparative anatomist and a specialist in the study of **vertebrate evolution**.

Unit

4

Animal Diversity - II

The 'Bioarchitecture' of the 'Chordates' and the 'Emerging Bioarchitect'

- A Brief History of 'Chordates'

About 60,000 known species of highly evolved, heterogeneous group of organisms constitute the phylum **Chordata**. The evolution of the chordates dates back to more than half a billion years. The development of a flexible rod like '**notochord**' along the dorsal aspect of the embryo and the **attachment of various muscles to that 'endoskeletal structure'** helping side to side 'swinging movements' can be considered a major factor in the evolution of chordates. Chordates are '**enterocoelous deuterostomes**', with their closest invertebrate relatives in the group **Echinodermata**.

A mid dorsal, ectodermal, hollow, non-ganglionated 'nerve cord' (above the notochord), which differentiates into the '**brain**' and '**spinal cord**'- is another contrasting feature, when compared to an invertebrate (non-chordate). The notochord is replaced by a '**vertebral column**', which encloses and protects the nerve cord, in the higher chordates (vertebrates/craniates). There are two more distinguishing features-the **pharyngeal slits** and **muscular post-anal tail**. The urochordates and the cephalochordates are the **non-vertebrate chordates**. The appearance of 'jaws' is perhaps the **major evolutionary step** that led to the domination of the 'jawed fishes' during the **Devonian Period**. The amphibians **invaded land**, but they remained an '**imperfect group**' as they could not adapt completely to terrestrial life. They have to return to water to lay eggs. The next group to have evolved were the reptiles and they (the giant bodied dinosaurs) dominated the Earth, during the **Mesozoic Era**. They developed a **dry scaly skin** and '**cleidoic eggs**' (also called '**amniotic eggs**'). The reptiles gave rise to two major groups – the **AVES** and the **MAMMALIA**. Kidneys

evolved from the primitive '**pronephros**' to '**mesonephros**' and finally the '**metanephros**'. There was a continuous complexity in the development of the '**brain**'. Fossil records of **connecting links** such as the '**osteolepid fishes**', '**labyrinthodont amphibians**', '**therapsid reptiles**' and '**theropod dinosaurs**' are available to **endorse** the presumed course of chordate evolution. The origin of **amnion** and other foetal membranes in the reptiles, **air sacs**, **feathers** of *birds*, the **hair**, **mammary glands** and **chorio-allantoic placenta** of mammals are the major '**Bioarchitectural wonders**', the chordate world witnessed through the course of evolution. The current ERA is definitely the '**AGE OF THE MAMMALS**'. *Homo sapiens sapiens* the 'modern man' is sitting on the top, 'visualising' the ENTIRE HISTORY OF EVOLUTION on his '**mind's screen**'. There was a tremendous '**genetic transformation**' making man **unique**. For example, the gene **FOX P2** is believed to play a key role in **human language expression** (along with some other essential genes) that kept man-**apart from the other organisms**. They enabled him develop **cognitive skills**, richly deserving his specific epithet '**sapiens**' (meaning '**wise**'/'**intelligent**')- a name which was so optimistically given by **Linnaeus**. Well, we had a look into the **bioarchitecture** of **chordate body** and **who is this emerging architect?** Ofcourse **MAN** himself, because he is guiding the process of evolution – **making himself the '**EMERGING BIOARCHITECT**'**.

- 4.0 Phylum – Chordata
- 4.1. Subphylum – Urochordata
- 4.2. Subphylum – Cephalochordata
- 4.3. Subphylum – Vertebrata
- 4.4. Super class – Agnatha
 - 4.4.1. Class – Cyclostomata
- 4.5. Super class – Gnathostomata
 - 4.5.1 – Pisces
 - 4.5.2. Class – Chondrichthyes
 - 4.5.3. Class – Osteichthyes
- 4.6. Tetrapoda
 - 4.6.1. Class – Amphibia
 - 4.6.2. Class – Reptilia
 - 4.6.3. Class – Aves
 - 4.6.4. Class – Mammalia

4.0 PHYLUM: CHORDATA

Chordates are one of the most familiar groups of animals to mankind. Phylum **Chordata** includes all those animals that possess the notochord (a primary endoskeletal element, their **chief distinctive** character) at some stage in their life time. It is one of the major animal phyla in terms of the number of species. Chordates are believed to have descended from **echinoderm larvae-like ancestors** during the **Precambrian** period. Chordates include **protochordates** (tunicates, lancelets) and **vertebrates** (cyclostomes, fishes, amphibians, reptiles, aves and mammals). Most of them are aquatic and some are amphibious or terrestrial animals. Though they exhibit variation in the size and shape of their bodies, they display similarity in their basic body plan. The first and the second largest of the living animals, namely blue whale (***Balaenoptera musculus*** - a marine mammal) and whale shark (***Rhinodon typus***) are members of the phylum Chordata.

I. The four principal chordate characters

1. Notochord

It is a flexible rod like structure situated along the mid dorsal line between the gut and the nerve cord. It is the first part of the endoskeleton to appear in the embryo. It is derived from the embryonic **chorda mesoderm**. It is made up of a core

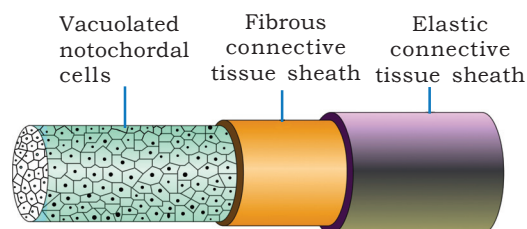


Fig. 4.1 Longitudinal view of notochord

of 'vacuolated cells' surrounded by an inner thick **fibrous** and an outer thin **elastic** connective tissue sheaths. It is persistent throughout their life in the **lancelets**, **cyclostomes** and fishes of the group chondrichthyes (NCERT). It is present in the tail of the tadpole larva of an **ascidian** which is lost in its adult stage due to **retrogressive metamorphosis** (see Glossary). It is present in the embryonic stages, but is replaced partly or wholly by the vertebral column in the adults of higher chordates. Remnants of notochord occur as **nuclei pulposi** in the **intervertebral discs** of mammals.

2. Dorsal tubular nerve cord

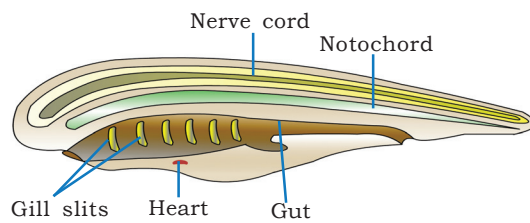


Fig. 4.3 Typical chordate

A single, hollow tubular and fluid filled nerve cord is situated above the notochord and below the dorsal body wall. It is **non-ganglionated**, unlike that of the non-chordates. It is produced in the embryonic stage by the 'sinking in' of the median dorsal strip of **ectoderm** above the **notochord**. In the higher chordates, it gets enlarged to form a distinct **brain** at the anterior end and the rest of it becomes the **spinal cord**.

3. Pharyngeal slits or clefts

Pharyngeal slits are a series of lateral perforations in the wall of the pharynx through which water flows out from the pharyngeal cavity. They are **ecto-endodermal** in origin. They are persistent throughout the life in the protochordates, fishes and some amphibians. The walls of the pharyngeal slits develop vascular lamellae to become 'gills' (branchiae) and are helpful in the exchange of respiratory gases. They are present in the larval stages, but are absent in the adults of many amphibians. Among the amniotes, non-functional pharyngeal pouches appear in the early embryonic life and disappear later. The occurrence of such structures in the early embryos of amniotes provides a clue to their 'aquatic ancestry'.

4. Post-anal tail

Chordates have a tail extending posterior to the anus. It is lost in many species during the late embryonic development. It contains skeletal elements and muscles. However, coelom and visceral organs are absent in it. It provides the propelling force in the locomotion of many aquatic species and acts as a 'balancing organ' in some terrestrial forms such as Kangaroos.

Chordates share **deuterostomeate condition**, **radial** and **indeterminate** cleavage and **enterocoelom** with the echinoderms.

II. Other chordate characters

Bilateral symmetry, triploblastic condition, organ-system level of organisation, metamerism, cephalisation, 'enterocoelic' coelom (except in the higher chordates in which it is secondarily schizocoelic), 'complete' digestive tract, closed circulatory system, ventral 'myogenic' heart (except in the lancelets), phosphocreatine (muscle phosphagen), etc., constitute the other important features of the chordates.

Phylum Chordata is divided into three subphyla: **Urochordata** or **Tunicata**, **Cephalochordata** and **Vertebrata**. Urochordates and Cephalochordates are together referred to as '**protochordates**'.

III. Comparison of Chordates and Non-chordates

S.No.	Chordates	Non-chordates
1.	Notochord is present.	Notochord is absent.
2.	Central nervous system is dorsal, hollow, single and non-ganglionated.	Nerve cord is ventral, solid, double and ganglionated.
3.	Pharynx is perforated by gill slits.	Pharyngeal gill slits are absent.
4.	Heart is ventral.	Heart is dorsal (if present).
5.	A post-anal tail is present.	Post-anal tail is absent.

4.1 SUBPHYLUM-UROCHORDATA OR TUNICATA

(**GR.OURA-** A TAIL; **L. CHORDA-** CORD)

All urochordates are marine and occur from the surface water to greater depths. They are either sessile (ascidians) or pelagic (*Salpa*, *Doliolum*) and solitary (*Ascidia*) or colonial (*Pyrosoma*). Body is un-segmented and covered by a **test** or **tunic** composed of **cellulose**¹ which is uncommon in animals. Coelom is absent. However, an ectoderm-lined **atrial cavity**, surrounds the pharynx, into which gill slits, anus and genital ducts open. Ventral side of the pharynx possesses an **endostyle**, which is believed to be the forerunner of the thyroid gland of a vertebrate (see Glossary). Atrium leads to the exterior by a dorsal or posterior atrial aperture. Digestive tract is complete. Two to numerous gill slits are found in the pharyngeal wall. Circulatory system is of **open type**. There is a simple, tubular, **ventral heart** which alternately reverses the direction of the flow of blood. Nervous system is represented in the adult by a single dorsal 'ganglion'. They are **bisexual** or **hermaphrodites**. Development generally includes a free-swimming **tadpole larva** with a tail, a dorsal hollow nerve cord, and a notochord confined to the tail, hence the name **urochordata**.

e.g. *Ascidia*, *Salpa*, *Doliolum*, *Pyrosoma* and *Oikopleura*.

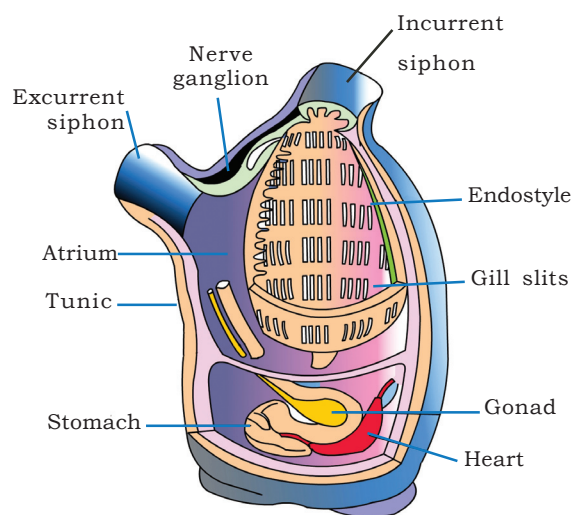


Fig. 4.4 *Ascidia*– Internal organisation

Is it necessary to know the life history of an ascidian to consider it a chordate?

¹ Ref. Integrated Principles of Zoology 12th Edition by Hickman and Biology 9th Edition by Raven and Johnson.

4.2 SUBPHYLUM-CEPHALOCHORDATA

(Gr. *KEPHALE*- HEAD; L. *CHORDA*- CORD)

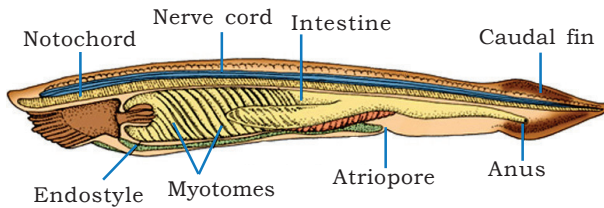


Fig. 4.5 *Branchiostoma*

Cephalochordates are marine animals which lead a burrowing mode of life in shallow sea waters. They are small 'fish like', translucent forms with median fins but without paired fins. They are often described as **typical chordates**, because they possess the principal chordate characters such as **notochord**, **tubular nerve cord** and

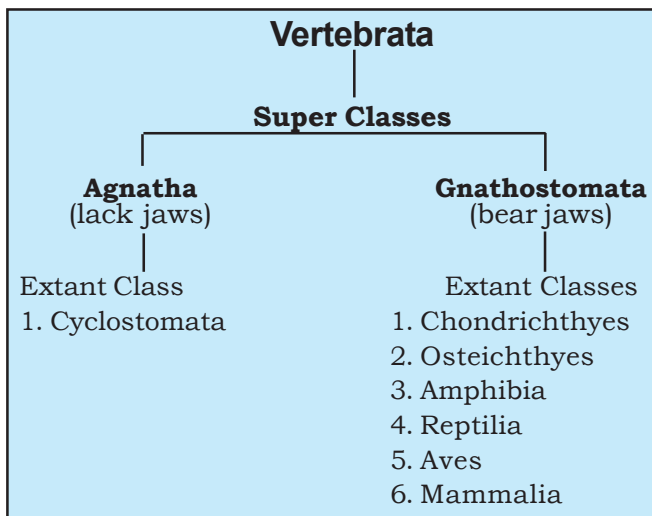
pharyngeal slits throughout their life. The notochord extends from the posterior end to the anterior end of the body beyond the nerve cord. Muscles occur as **myotomes** (blocks of muscles). The coelom is **enterocoelic**. They are **ciliary** or **filter feeders**. The pharynx is surrounded by an ectoderm lined cavity called **atrium**. An **endostyle** is present along the ventral wall of the pharynx. Exchange of respiratory gases occurs mostly across the **external body surface**. Circulatory system is of **closed type** and heart, blood corpuscles and respiratory pigment are absent. Excretion is performed by solenocytes of protonephridiae type. Numerous gonads without gonoducts are present in the pharyngeal region. **Fertilization is external** and development is **indirect** (involves a free swimming larval form).

e.g. *Branchiostoma* (amphioxus or lancelet)

Branchiostoma is often described as a typical chordate. Justify

4.3 SUBPHYLUM-VERTEBRATA/ CRANIATA

Classification of subphylum Vertebrata



The members of the subphylum Vertebrata possess notochord during the embryonic period which is replaced partly or wholly by a cartilaginous or bony vertebral column in the adult. It is true that "**all vertebrates are chordates but all chordates are not vertebrates**". Besides the basic chordate characters, vertebrates have paired appendages which may be 'fins' or 'limbs' and they are with or without jaws. Heart is ventral, muscular with two, three or four chambers. **Mesonephric** or **metanephric** kidneys are the organs of excretion and osmoregulation.

4.4 SUPER CLASS: AGNATHA (Gr. A- WITHOUT; GNATHOS- JAW)

They are jawless primitive fish like aquatic vertebrates without paired appendages. The internal ear bears one or two semicircular canals. Notochord persists in the adult.

Agnatha includes the extant class **Cyclostomata** besides the extinct class **Ostracodermi**.

4.4.1 Class: Cyclostomata (Gr. *cyklos*- circular; *stoma*- mouth)

Cyclostomes are jawless aquatic (marine or freshwater) forms. Body is scaleless, long, slender and eel-like. Endoskeleton is cartilaginous. Vertebrae are represented by imperfect neural arches in some. Mouth is circular and suctional. Tongue bears **horny teeth**. Respiratory organs include six to fifteen pairs of gill slits. Heart is **two-chambered** as in fishes. **Renal portal system is absent**. Kidneys are **mesonephric**.

e.g. *Petromyzon* (lamprey) and *Myxine* (hag fish/slime eel).

Marine **lampreys** migrate to freshwater for spawning (anadromous migration). After spawning they die. Their larvae are **ammocoete** metamorphose into adults and return to the sea.

Hagfishes are exclusively marine. They are considered scavengers because they feed on dead or dying fish. When disturbed hag fish releases a milky fluid from its mucous sacs which becomes slime on contact with sea water. Development is direct.



Fig. 4.6 *Petromyzon*

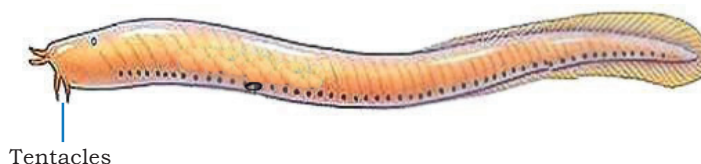


Fig. 4.7 *Myxine*

4.5 SUPER CLASS: GNATHOSTOMATA (Gr. GNATHOS- JAW; STOMA- MOUTH)

They are vertebrates with jaws. They possess paired appendages (pectoral and pelvic fins or fore and hind limbs) and internal ears with three semicircular canals each. Notochord is replaced partly or wholly by the vertebral column.

Gnathostomata includes **jawed fishes** (Pisces) and **Tetrapods** (Amphibia, Reptilia, Aves and Mammalia).

- Pisces and Amphibia constitute the group **Ichthyopsida**
- Reptilia and Aves constitute the group **Sauropsida**.

4.5.1 Pisces (L. *piscis*- fish)

Fishes, the 'first jawed vertebrates', are completely aquatic and gill breathing animals. The extinct jawless fishes, the **ostracoderms** gave rise to the earliest jawed fishes during the **Silurian** period. **Devonian** period is considered the '**Golden age of fishes**', because they emerged as the predominant group during that time. They constitute the most **diverse** and **largest** vertebrate group. They show variation in their body size, shape, and colour. They are one of the economically most important animal groups to mankind. **Coelacanth**s (marine) and **Dipnoi** (fresh water) are the extant 'lobe finned' fishes. **Sharks**,



Fig. 4.8 *Latimeria chalumnae*

rays and **ray-finned fishes** account for the bulk of living fishes. Osteolepid fishes which are of the same group as *Latimeria* are believed to be in the line of evolution of the next higher group of vertebrates, the amphibians. The study of fishes is known as **Ichthyology**.

Name the fish which was thought to be extinct until 1938, but later found to be living in small numbers.

General characters

- ❖ Fishes are completely aquatic poikilothermic (cold blooded) animals.
- ❖ Body of a fish is usually streamlined and differentiated into head, trunk and tail.
- ❖ Exoskeleton consists of **mesodermal** scales or bony plates. A few are scaleless.
- ❖ Endoskeleton may be cartilaginous or bony. Skull is **monocondylic**. Vertebrae are **amphicoelous** (centrum is concave at both anterior and posterior faces).
- ❖ Locomotion is assisted by unpaired (median and caudal) fins along with paired (pectoral and pelvic) fins.
- ❖ Mouth is **ventral** or **terminal**. Teeth are usually **acrodont**, **homodont** and **polyphyodont**.
- ❖ Exchange of respiratory gases is performed by the **gills**. Heart is 'two chambered' and is described as **branchial heart** as it supplies blood only to the gills. The circulation is single circulation as blood reaches the heart only once, in the course of each circulation, making the heart a **venous heart** (heart receives only venous/deoxygenated blood from the body parts).

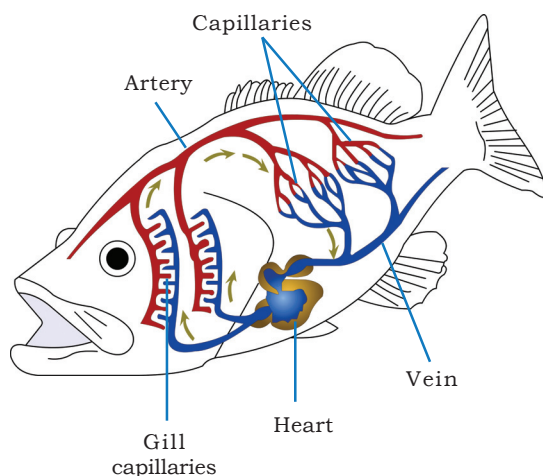


Fig. 4.9 Circulation of blood in a fish- schematic representation

- ❖ Kidneys are **mesonephric**. Fishes are mostly **ammonotelic** and some are **ureotelic** (cartilaginous fishes).
- ❖ Cranial nerves are 10 pairs. **Meninx primitiva** is the only 'meninx' enveloping the central nervous system.
- ❖ Internal ear consists of three semicircular canals. **Lateral-line sensory system** (to detect movement and vibration in the surrounding water) is well-developed. Eyes are without eyelids and each eye ball is protected by a **nictitating membrane**.
- ❖ Sexes are separate. Fertilization is internal or external. Development may be direct or indirect.

Pisces include two extant classes i.e. **Chondrichthyes** and **Osteichthyes**.

4.5.2 Class - Chondrichthyes / Cartilaginous fishes)

(Gr. **chondros**- cartilage; **ichthys**- fish)

They are marine fishes which possess cartilaginous endoskeleton. Caudal fin is **heterocercal** (asymmetrical both externally and internally). Skin is covered by dermal **placoid** scales also called dermal denticles. Sharks are highly predaceous. Respiratory gaseous exchange is performed by five to seven **lamelliform** gills, without operculum, on each side. Air bladder is absent, and so they have to swim constantly to avoid sinking (bony fishes have 'air bladder' acting as a 'hydrostatic organ' helping the fish float easily at the desired level without much expenditure of energy). Cartilaginous fishes are **ureotelic** and store urea in their blood to maintain osmotic concentration of body fluids (physiological uraemia). In males, pelvic fins bear **claspers** to facilitate internal fertilization. They are mostly **viviparous**.

e.g. *Scoliodon* (dog fish), *Pristis* (saw fish), *Carcharodon* (great white shark), *Dasyatis* / *Trygon* (sting ray; possesses poison sting), *Torpedo* (electric ray; dorsal muscles are modified into electric organs), *Sphyrna* (hammer - headed shark).

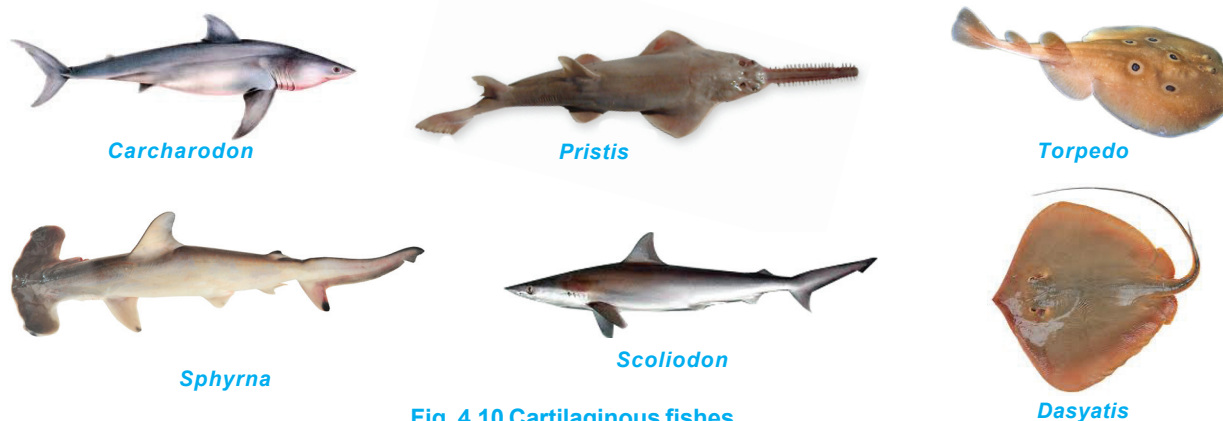


Fig. 4.10 Cartilaginous fishes

Why are sharks ureotelic, while most of the bony fishes are ammonotelic?

4.5.3 Class - Osteichthyes (Gr. *osteon*- bone; *ichthys*- fish)(Bony fishes)

They are bony fishes which live in all kinds of aquatic habitats. Caudal fin is **homocercal** (symmetrical externally and asymmetrical internally) in **teleost fishes** and **diphycercal** (symmetrical both externally and internally) in lung fishes and *Latimeria*. Exoskeleton is in the form of **cosmoid**, **ganoid**, **cycloid** or **ctenoid scales**. Mouth is usually **terminal**. Respiratory gaseous exchange is performed by four **filamentous gills** covered by an **operculum**, on each side. An air bladder is present with or without connection to the gut. It is either helpful in gas exchange (lung fishes) or in maintaining buoyancy (hydrostatic function) in most of the 'ray-finned' fishes. They are mostly **ammonotelic**. Sexes are separate. Fertilization is generally external. Most forms are **oviparous**.

Examples: Marine fishes: *Exocoetus* (flying fish), *Hippocampus* (sea horse; male has a brood pouch), *Echeneis* (sucker fish); Fresh water fishes: *Catla* (katla), *Labeo* (Rohu), *Clarias* (magur); Aquarium fishes: *Betta splendens* (Siamese fighting fish), *Pterophyllum* (angel fish).

"Teleostians are the most successful group of fishes". Explain why?

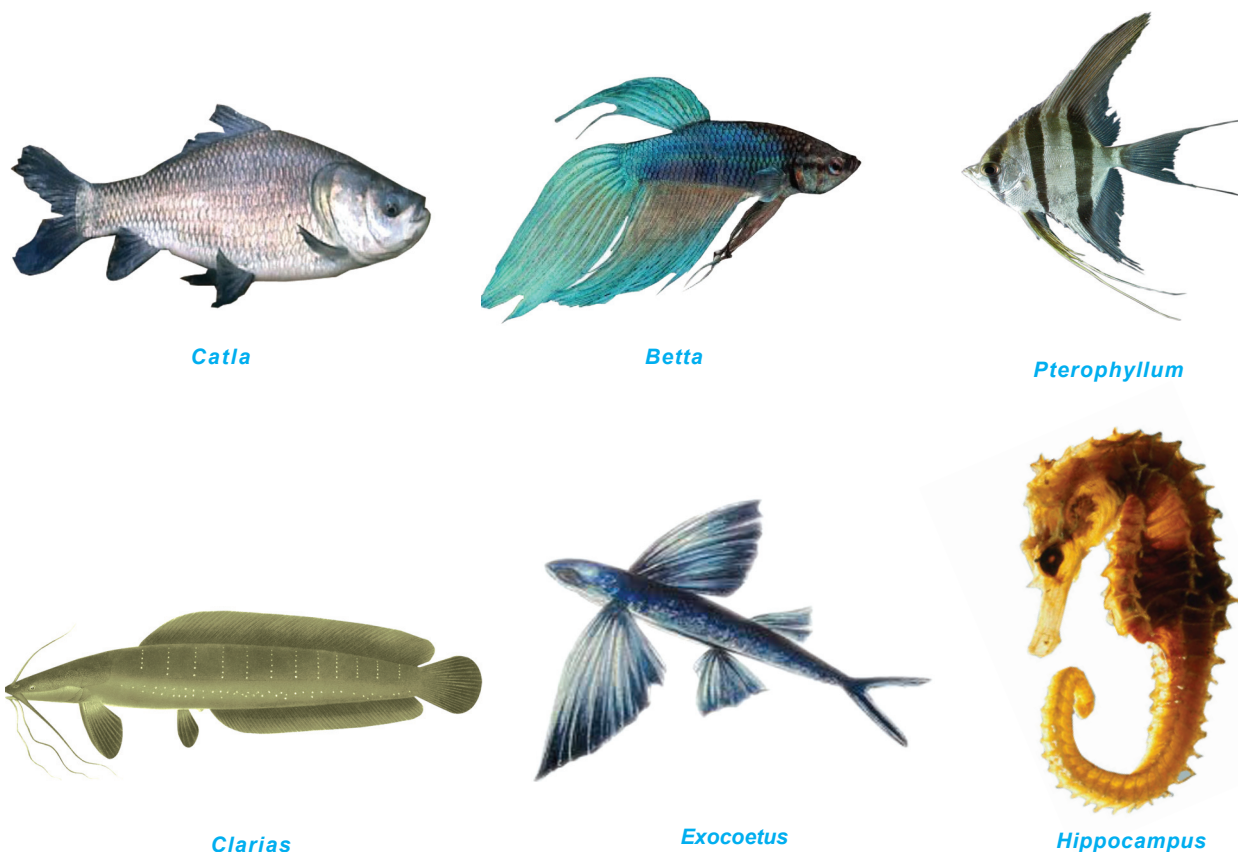


Fig. 4.11 Bony fishes

4.6 TETRAPODA (GR. *TETRA*-FOUR; *PODOS*-FOOT)

They are mostly terrestrial but some are **aquatic** or **amphibious** animals. They are characterised by the presence of two pairs of **pentadactyl limbs** usually. Lungs are the principal organs of respiratory gas exchange (in most of the amphibians, skin is the principal site of gaseous exchange).

Tetrapoda includes four classes: **Amphibia**, **Reptilia**, **Aves** and **Mammalia**.

4.6.1 Class: Amphibia (Gr. *amphi*-both; *bios*-life)

Amphibians are the first vertebrates to come out of water and 'walk' on land. They are the tetrapods which have the ability to live in the aquatic habitat of their ancestors, the fishes (osteolepids) and in the terrestrial habitat that they

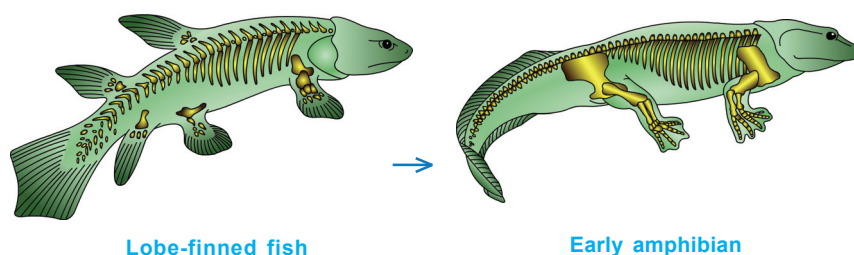


Fig. 4.12 Origin of an amphibian

first invaded. They flourished during the **carboniferous** period. They could not become completely terrestrial animals as they are still tied to **aquatic habitat** for **reproduction** and **development** (because the eggs of amphibians dry out quickly if laid on land and they do not have a 'shell' to suit the hard surface on land). The living amphibians are **anurans** or **salientians** (frogs and toads), **urodeles/caudates** (salamanders and newts) and **apodans** (caecilians/blind worms). The study of amphibians is known as **batrachology**.

I. General characters

- ❖ They are the **first tetrapods** and lead a **dual mode of life**, i.e. on land and in freshwater.
- ❖ Body is divided into distinct '**head**' and '**trunk**'. Tail may or may not be present.
- ❖ Skin is **soft**, **scale-less** (except the members of Apoda), **moist** and **glandular**.
- ❖ The body bears two pairs of **equal** or **unequal pentadactyle** limbs (caecilians are limbless)
- ❖ Skull is **dicondylic** as in mammals. Vertebrae are mostly **procoelous** (centrum is concave at its anterior face only) in the anurans, **amphicoelous** in the caecilians and usually **opisthocoelous** (centrum is concave at its posterior face) in the urodeles. **Sternum** appeared for the first time in the amphibians.
- ❖ Mouth is large; teeth are **acrodont**, **homodont** and **polyphyodont**.
- ❖ Respiratory gaseous exchange is mostly **cutaneous**; **pulmonary** and **bucco-pharyngeal** respirations also occur. Branchial respiration is performed by larvae and some adult urodeles.

- ❖ Heart is three-chambered with **sinus venosus** and **conus arteriosus**². Three pairs of aortic arches and well-developed portal systems are present; erythrocytes are nucleate.
 - ❖ Kidneys are **mesonephric**; adult amphibians are **ureotelic**.
 - ❖ Meninges are the inner **piamater** and outer **duramater**; cranial nerves are 10 pairs.
 - ❖ Middle ear consists of a single ear ossicle, the **columella auris** which is the modified '**hyomandibula**' of the fishes. **Tympanum**, **lacrimal** and **harderian glands**³ appeared for the first time in the amphibians.
 - ❖ Sexes are separate and fertilization is mostly external. Development is mostly indirect.
- e.g. *Ichthyophis* (limbless amphibian), *Bufo* (toad), *Rana* (frog), *Hyla* (tree frog), *Rhacophorus* (flying frog), *Salamandra* (salamander).

There are no amphibians which are completely adapted to terrestrial environment with reference to completing their life cycle. Justify.



Ichthyophis



Salamandra



Rhacophorus

Fig. 4.13 Amphibians

II. Frog

Frog and toad are tailless amphibians that belong to the order **Anura/ Salientia**.

1. Habitat and Habit: They live in a variety of environments. Most of the frogs live in or near water, whereas the toads are well-adapted to comparatively dry environments. All the adult frogs and toads are **carnivores**. *Rana tigrina* is the most common Indian frog which is popularly known as **Indian bull frog**. Frogs and toads are **poikilothermic anamniotes** (cold-blooded animals and do not develop amnion).

2. Hibernation and Aestivation: During adverse environmental conditions such as severe winter or dry hot summer season, frogs bury themselves in the soft and damp mud, to tide over the adverse climatic conditions. They slow down their metabolism during those periods. They survive on glycogen and fat reserves. They perform only cutaneous respiration during those periods. The period of inactive life during winter is

² Ref. Integrated Principles of Zoology 12th Edition by Hickman

³ Lacrimal and Harderian glands are associated with the eyes.

called **hibernation** or 'winter sleep' and the period of inactive life during summer is called **aestivation** or 'summer sleep'.

3. Camouflage: Frogs are not easily spotted by their enemies as they can change the colour of their skin to match with that of their surroundings. They have the ability to change their colour to hide from enemies and it is called camouflaging.

III. *Rana tigrina*

Class : Amphibia
Order : Anura/ Salientia
Family : Ranidae

The body of frog is somewhat spindle shaped and slightly flattened dorsoventrally. Skin of frog is **thin**, **scaleless** and kept **moist** by the mucus glands. It is attached loosely to the underlying muscles at certain points only. The skin on the dorsal surface is generally olive-green with irregular dark spots. On the ventral surface, the skin is uniformly pale yellow. Frog never drinks water but absorbs it through the skin.

1. Body Divisions

The body of a frog is divisible into **head** and **trunk**. The neck and tail are absent. Above the snout, a pair of external nostrils is present. The eyes are bulged and covered by a nictitating membrane that protects them while in water. Behind each eye, a membranous tympanum (ear drum) receives sound waves. The forelimbs and the hind limbs help in swimming, walking, leaping and burrowing. The hind limbs bear five digits each. They are larger and more muscular than the forelimbs, which bear four digits. Feet have webbed digits and the web helps in swimming.

Frog exhibits sexual dimorphism. Male frog can be distinguished by the presence of sound amplifying **vocal sacs** and also a **copulatory pad** on the first digit of each forelimb (copulatory pads are absent in female frogs).

2. Coelom and Viscera

The body cavity of frog accommodates different internal organs or **viscera**, which are covered by visceral peritoneum.

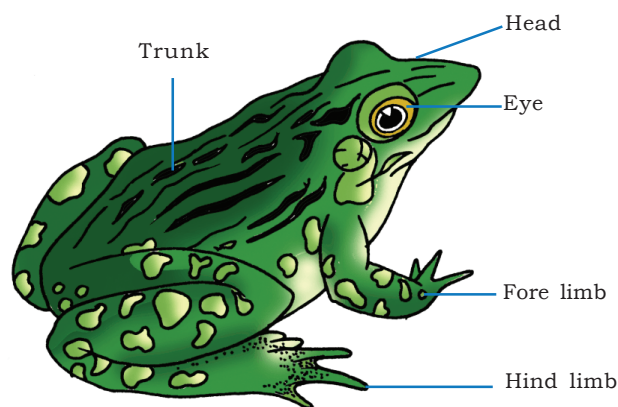


Fig. 4.14 External features of frog

3. Digestive System

The digestive system consists of **alimentary canal** and **digestive glands**. The alimentary canal is short, because frog is a carnivore.

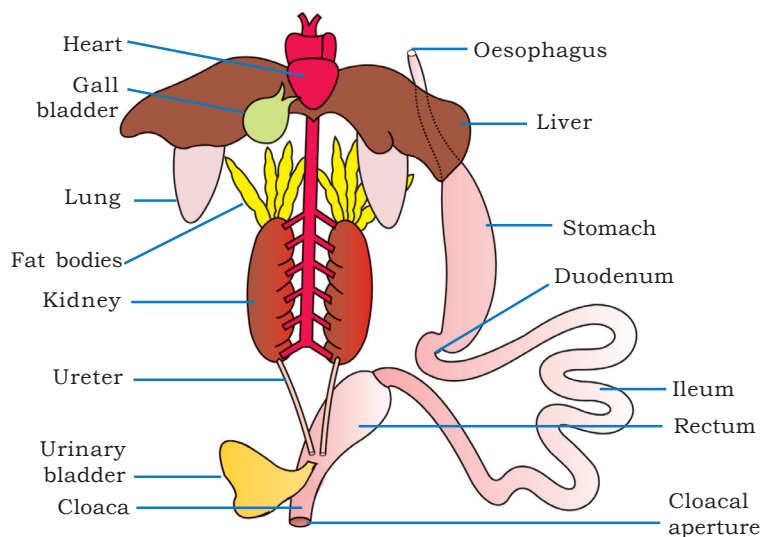


Fig. 4.15 Diagram showing digestive system and some visceral organs

are provided with **cardiac** and **pyloric sphincters** respectively. The stomach is an organ for the **storage** and **digestion of food**. The small intestine consists of the proximal part, the **duodenum** and the distal long and coiled part, the **ileum**. The ileum opens into the **rectum** which is also called the **large intestine**. It is a short, wide tube that opens into the cloaca through the **anus**, which is guarded by the anal sphincter. Cloaca is a small chamber, which serves as a common passage for the digestive and urinogenital systems. It opens out through the cloacal aperture at the posterior end of the trunk between the hind limbs.

B. Digestive glands

Gastric glands of the stomach are microscopic and secrete the gastric juice. **Liver** is the 'largest gland' and it opens into the duodenum through hepato-pancreatic duct as it collects pancreatic juice also on the way. It secretes bile, which is stored in the gall bladder. Bile has no digestive enzymes, but contains bile salts and pigments. **Pancreas** is an irregular, elongated **mixed gland**⁴, situated between the stomach and the duodenum. It produces pancreatic juice. **Intestinal glands** secrete the 'intestinal juice' (also called succus entericus).

A. Alimentary Canal

It begins with the mouth which is a wide, semi-oval slit bound by the upper and lower jaws. Mouth opens into a large, wide and shallow **bucco-pharyngeal cavity**. It shows teeth along the margin of the upper jaw, a muscular and bifid tongue which is attached in front and free behind. It opens into a short **oesophagus** which merges gradually with the **stomach**. The stomach is a broad, thick-walled tube that occupies the left side of the body cavity. The anterior and posterior ends of the stomach

⁴ Pancreatic islets of Langerhans secrete hormones such as insulin and glucagon.

C. Feeding and Digestion: The tongue is an organ of food capture. Digestion begins in the stomach by the action of gastric enzymes in the presence of HCl. Partly digested acidic food, called **chyme** is passed through the pylorus into the small intestine, where it is mixed with **bile juice**, **pancreatic juice** and **intestinal juice**. Bile emulsifies fats, while pancreatic and intestinal enzymes complete the digestion of carbohydrates, proteins and lipids. The digested food is absorbed through the 'villi' and microvilli in the inner wall of the intestine. The undigested solid wastes move into the rectum and pass out through the cloacal aperture via the anus and cloacal sac.

4. Respiratory System

Frog uses three types of respiratory surfaces. They are the skin, mucous lining of bucco-pharyngeal cavity and inner lining of the lungs.

- Cutaneous respiration:** It is the most important of the three types. The scaleless skin in frog is thin, **moist** and **vascular**. The skin is permeable to O_2 and CO_2 . The O_2 dissolved in mucous film/moisture on the surface of the body diffuses into the blood capillaries in exchange of CO_2 . Skin should be moist for cutaneous respiration.
- Bucco-pharyngeal respiration:** Respiratory exchange of gases occurs through moist, vascular lining of its bucco-pharyngeal cavity in addition to skin and lungs when it is on land.
- Pulmonary respiration:** When frog is on land it also breathes through its lungs. The lungs are pink coloured highly vascular sac-like organs. During the pulmonary respiration the bucco-pharyngeal cavity acts like a '**force pump**'. Due to the elevation of bucco-pharyngeal cavity the air forces through the glottis and enters the lungs, where exchange of gases takes place. After the exchange of gases, the floor of the bucco-pharyngeal cavity is lowered and the air from the lungs goes out via bucco-pharyngeal cavity and nostrils.

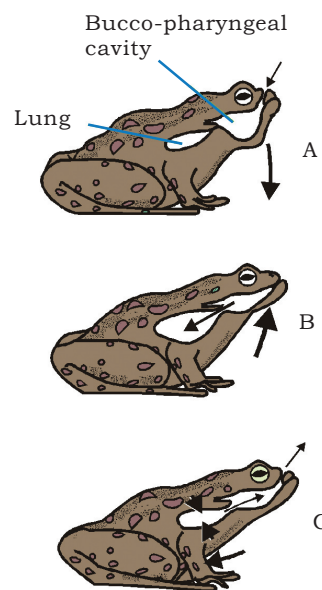


Fig. 4.16 Steps during pulmonary respiration

5. Blood Vascular System

The blood vascular system consists of the heart, blood vessels and blood. The heart is a muscular organ situated in the antero-ventral part of the body cavity. It has two separate atria and a single undivided ventricle. It is covered by a double layered membrane called **pericardium**. A triangular chamber called **sinus venosus** joins the right atrium on the dorsal side. It receives blood through three **vena cavae** (caval veins). The ventricle opens into the **conus arteriosus** (Ref: NCERT) on the ventral

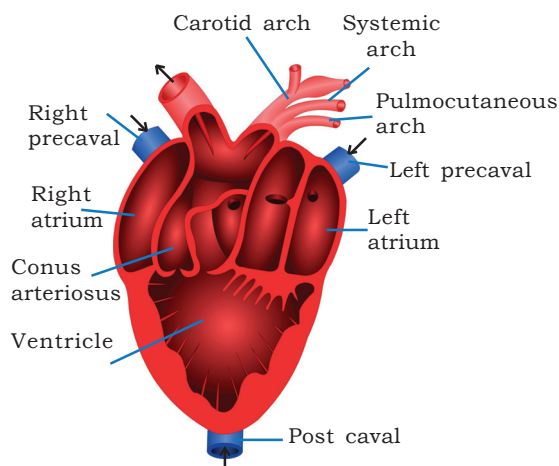


Fig. 4.17 Frog heart

side. The conus arteriosus bifurcates into two branches and each of it divides into three aortic arches namely **carotid**, **systemic** and **pulmocutaneous**. Blood from the heart is distributed to all parts of the body by the branches of aortic arches. Three major veins collect blood from the different parts of the body and carry it to the sinus venosus. Circulation is described as 'Incomplete Double' circulation.

Portal Systems: Special venous connections between the intestine and liver as well as lower parts of the body and kidneys are present in frog. These are called hepatic and renal portal veins, respectively.

Blood: Blood is composed of plasma and cells. The blood cells are **erythrocytes**, **leucocytes** and **thrombocytes**. RBCs are nucleate. WBCs are nucleated amoeboid cells which help in defence. Thrombocytes help in **haemostasis** (stopping of bleeding).

Lymphatic system: In addition to the blood vascular system, frog has a lymphatic system. The lymphatic system consists of **lymph**, **lymph channels** and **lymph nodes**. Lymph is different from blood. It lacks RBCs and contains fewer proteins, than those in the blood.

6. Integrating System

Control and co-ordination of body systems is highly developed in frog. It includes both nervous and endocrine systems. The chemical integration of various organs of the body is achieved by the hormones, which are secreted by the endocrine glands. The prominent endocrine glands found in frog are **pituitary**, **thyroid**, **parathyroid**, **thymus**, **pineal body**, **pancreatic islets**, **adrenal glands** and **gonads**.

The nervous system is organised into a **central** (brain and spinal cord), **peripheral** (cranial and spinal nerves) and **autonomous** (sympathetic and parasympathetic) nervous systems.

Brain is enclosed in a bony structure called brain box (cranium). The brain is divided into **fore brain**, **mid brain** and **hind brain**. The fore brain includes two **olfactory lobes** (concerned with sense of smell), two **cerebral hemispheres** (control of voluntary actions) and **diencephalon** (perception of heat, cold, pain and integrity of autonomous system).

Mid brain is represented by a pair of **optic lobes** (*corpora bigemina*). The **optic lobes** are associated with the *sense of sight*. The hind brain consists of **cerebellum** (maintains equilibrium) and **medulla oblongata** (controls involuntary movements). Medulla oblongata passes out through the foramen magnum and continues as spinal cord which is enclosed in the neural canal of the vertebral column. The spinal cord acts as the '**middle man**' between the **brain** and **effectors**.

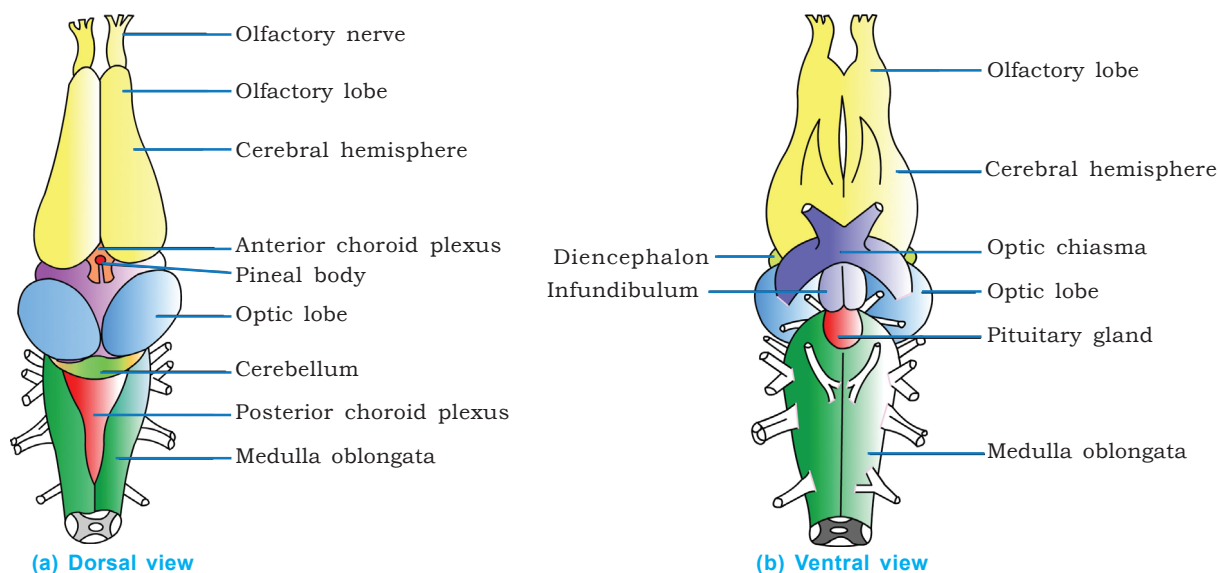


Fig. 4.18 Frog Brain

7. Special senses

Frog has sense organs such as the organs of **touch**, **taste**, **smell**, **sight** and **hearing**. The well-organised structures among them are eyes, internal ears and the rest are 'cellular aggregations' around **nerve endings**. The receptors of touch (sensory pupillae) occur in the **skin**. Organs of taste are called **taste buds** that lie on small papillae of tongue. The organs of smell are a pair of **nasal chambers**.

The organs of sight are a pair of **eyes** located in the orbits of the skull. Eyes are protected by eyelids. The upper eyelid is immovable. The lower eyelid is folded into a transparent **nictitating membrane**, which can be drawn across the surface of the eye. The **retina** of the eye contains both **rods** and **cones**. Cones provide 'colour vision' and rods are helpful in 'dimlight vision'.

Ear is useful for **hearing** and **balance**. It consists of a middle ear closed externally by a large **tympenic membrane** (ear drum) and a **columella** that transmits vibrations to the inner ear. The inner ear consists of a **utricle** with three **semicircular canals** and a small **sacculus**.

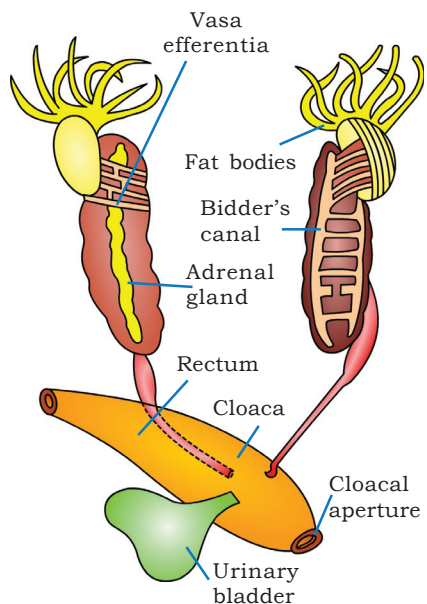
8. Excretory System

The elimination of the nitrogenous wastes and the maintenance of water and salt balance are performed by a well-developed excretory system. It consists of a pair of **kidneys**, **ureters**, a **urinary bladder** and a **cloaca**.

The kidneys are elongated, flat and dark red coloured structures situated a little posteriorly in the body, one on either side of the vertebral column. They are covered by peritoneum on their ventral (lower) surface only (retroperitoneal). An adrenal gland is found along the mid ventral surface of each kidney. The functional units of the kidney are **uriniferous tubules** or **nephrons**. From the posterior outer margin of each kidney,

a mesonephric or **Wolffian** duct arises. The ureters act as ‘**urino-genital ducts**’ in males because both urine and male gametes pass through them. However, in the females, the oviducts and ureters are separate. Ureters open into the cloaca by sphinctered apertures. A thin walled urinary bladder, present ventral to the rectum, opens into the cloaca. The chief nitrogenous waste in frog is urea, hence, it is said to be ‘**ureotelic**’.

Do you know - urea is mostly formed in the liver and excreted by the kidney.



4.19 Frog : Male reproductive system

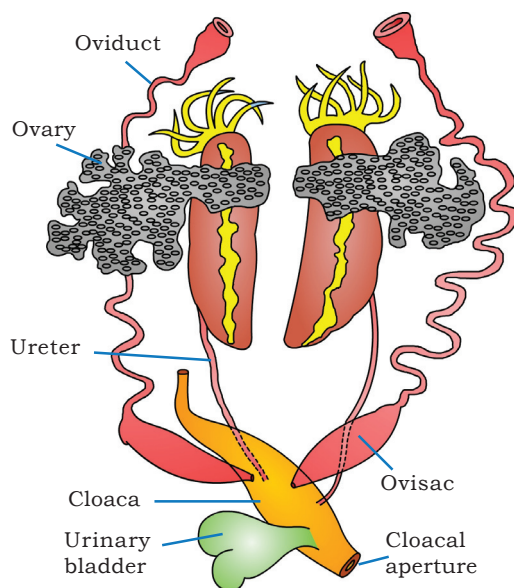


Fig. 4.20 Frog : Female reproductive system

9. Reproductive System

The male reproductive system consists of a pair of yellowish and ovoid testes, which are attached to the kidneys and dorsal body wall by a double fold of peritoneum called **mesorchium**. Each testis is composed of innumerable **seminiferous tubules** which are connected to form 10 to 12 narrow tubules, the **vasa efferentia**. They enter the kidneys and open into the **Bidder's canal** which is connected to the ureter through transverse canals of the kidney. The urino-genital ducts of both the sides open into the cloaca.

The female reproductive system consists of a pair of ovaries, oviducts and cloaca. The ovaries are attached to the kidneys and dorsal body wall by a double fold of peritoneum called **mesovarium**. The ovaries have no functional connection with the kidneys. The oviducts are two long, white and convoluted tubes. Their anterior ends form funnel-like openings called **Ostia**. The posterior ends of oviducts enlarge into **ovisacs** before they open into the cloaca separately.

During **amplexus**, the mass of eggs and the mass of sperms released by the female and the male are called **spawn** and **milt** respectively. Fertilization is external and takes place in water. Development involves an aquatic gill breathing, herbivorous larval stage called **tadpole**, which metamorphoses into an air breathing, carnivorous

adult frog. Tadpole of frog resembles a fish pertaining to taking in water and respiring through gills⁵.

Majority of the tetrapods possess an amnion during their development. Frogs are also tetrapods. Do their embryos too possess amnion? If not why?

10. Economic Importance of Frogs

Frogs are beneficial to mankind because they are insectivorous and help in reducing the insect population. In some countries such as China and Japan, the muscular thighs of frogs are used for human consumption.

4.6.2. Reptilia (*L. reptilis* - creeping)

Reptiles are ectothermic (organisms that regulate their body temperature largely by exchanging heat with their surroundings; cold-blooded) amniotes. The extinct **labyrinthodont** amphibians gave rise to reptiles during the carboniferous period. They emerged as the dominant vertebrate group during the **Mesozoic era** (Golden age of reptiles). Most reptiles lay water tight '**cleidoic**' eggs (covered by porous calcareous shell - to allow the passage of air to provide oxygen) on land with extra embryonic membranes namely **amnion**, **allantois**, **chorion** and **yolk sac**, which make the egg an independent 'life support system'. Thus, the '**amniotic egg**' helped the reptiles to abandon ties with their ancestral aquatic habitat. In addition to cleidoic eggs, the evolution of 'dry scaly skin' to prevent water loss, two pairs of pentadactyl limbs with clawed digits for moving on land, pulmonary breathing and internal fertilization are the key adaptations that led to the success of reptiles. The class reptilia includes the extinct **dinosaurs** (terrible lizards) and the extant **chelonians**, **crocodilians**, **snakes**, **lizards** and **Sphenodon**. The study of poikilothermic tetrapods is called **herpetology**.

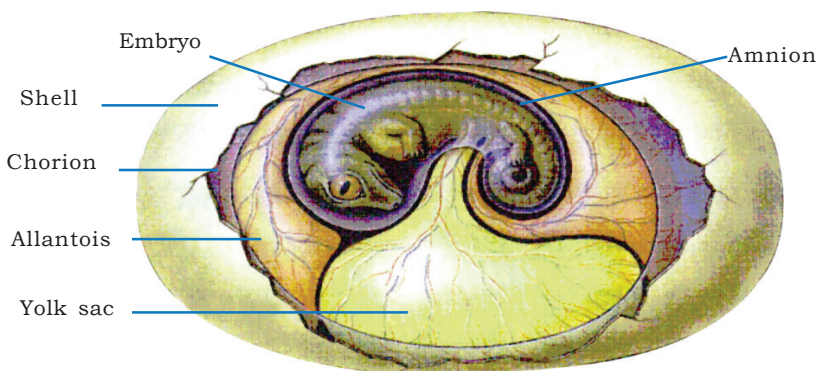


Fig. 4.21 Amniotic egg

What changes in the design of egg and embryo allowed reptiles to lay eggs on land? Why are their eggs often called 'amniotic eggs'?

⁵ Haeckel proposed that organisms during their embryonic development repeat/recapitulate some of their ancestral characters, even though they are non-functional. This is referred to as BIOGENETIC LAW/THEORY OF RECAPITULATION. It states 'ontogeny repeats phylogeny' (ontogeny: developmental history, phylogeny: ancestral history)

General Characters

- ❖ They are the first true terrestrial, usually creeping or burrowing, amniotic tetrapods.
- ❖ Body is divided into **head, neck, trunk** and **tail**.
- ❖ Skin is rough and dry. The exoskeleton occurs in the form of horny epidermal **scales, shields**, and **claws** (which appeared for the first time in reptiles).
- ❖ Dentition is **acrodont, homodont** and **polyphyodont** (thecodont in crocodiles as seen in the mammals). Chelonians are 'edentate'.
- ❖ Skull is **monocondylic** and many have **temporal fossae**. Each half of the lower jaw is formed by **six bones**. Vertebrae are mostly **procoelous**. The first two cervical vertebrae are specialized into **atlas** and **axis** to facilitate independent movement of the head from the rest of the body; sacral vertebrae are **two** in number.
- ❖ The exchange of respiratory gases takes place only through lungs. They use **ribs** and **intercostal muscles** in ventilation (this feature developed in the reptiles for the first time in the vertebrate evolution). In turtles, gaseous exchange takes place through the **vascular cloacal wall**.
- ❖ Heart is incompletely four-chambered, except in the crocodiles, in which the heart is four-chambered. **Sinus venosus** is present but **conus arteriosus** is absent. The three aortic arches arise directly from the ventricle. Erythrocytes are nucleate.
- ❖ Kidneys are **metanephric. Mesonephric duct** (Wolffian duct) of the embryo functions as 'vas deferens' in the males. They are '**uricotelic**' (uricotelism is an adaptation to conserve water).
- ❖ Cranial nerves are 12 pairs, except in the snakes (10 pairs)
- ❖ **Tympanic membrane** is found at the inner border of the external auditory meatus. Middle ear has a single ear ossicle called **columella auris. Jacobson's organs**, the specialized olfactory structures, are highly developed in **lizards** and **snakes**.
- ❖ Males usually possess a muscular copulatory organ, except in the rhynchocephalians. Lizards and snakes have a pair of **hemipenes** (penis is formed by the apposition of the two hemipenes).
- ❖ Cloaca is three-chambered as seen in birds. They are the anterior **coprodaeum**, middle **urodaeum** and posterior **proctodaeum**.
- ❖ Fertilization is internal; mostly oviparous (some snakes are viviparous). Eggs are **megalecithal** and **cleidoic**. Cleavage is **meroblastic⁶** and **discoidal**. During development, extra embryonic membranes are formed as in birds and mammals.

In what ways are the reptiles structurally and functionally more suited for the terrestrial mode of life than their predecessors?

⁶ The amount of the yolk in the egg influence the mode of cleavage.

The extant reptiles are grouped into four orders

1. **Chelonia** - *Chelone* (marine green turtle), *Testudo* (terrestrial form), *Trionyx* (fresh water form)
2. **Rhynchocephalia** - *Sphenodon* (a 'living fossil', endemic to New Zealand)
3. **Crocodylia** - *Crocodylus palustris* (Indian crocodile or maggur), *Alligator* (alligator), *Gavialis gangeticus* (Indian gavial or gharial)
4. **Squamata**
 - a) Lizards : *Hemidactylus* (wall lizard), *Chameleon*, *Draco* (flying lizard)
 - b) Snakes
 - i) **Poisonous snakes**: *Naja naja* (cobra), *Ophiophagus hannah* (King cobra), *Bungarus* (krait), *Daboia/Vipera russelli* (chain viper)
 - ii) **Non-poisonous snakes**: *Ptyas* (rat snake), *Tropidonotus* (grass snake or pond snake)

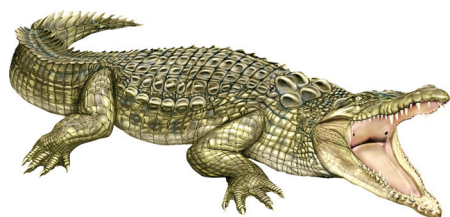
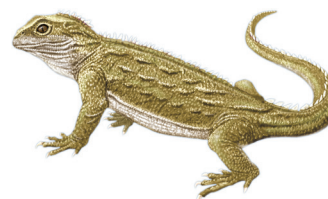
*Naja naja**Chameleon**Chelone**Crocodylus**Ptyas**Sphenodon***Fig. 4.22 Reptiles**



Fig. 4.23 Theropod reptile
(Imaginary diagram)

4.6.3 Class - Aves (L. avis-bird)

The class Aves includes a few extinct and a large number of extant birds. They are feathered, bipedal endothermic vertebrates. Modern flying birds have undergone modifications in their morphological, anatomical and physiological features as adaptations to suit their aerial mode of life. Feathers, wings, powerful breast musculature, **pneumatic bones**, endothermy with high metabolic rate, keen sense of vision, etc., are the major 'flight adaptations' which enabled them to evolve as '**the masters of air**' (J. Z. Young).

The presence of typical reptilian characters such as epidermal **scales on legs**, **presence of interclavicle**, **uricotelism**, **megalecithal eggs**, **development with the four extra embryonic membranes** in modern living birds, convinced **T. H. Huxley** to describe them as '**Glorified reptiles**'. The demands of flight paved the way for the evolution of a similar general body form in many flying birds. **Theropod reptiles** of the **Jurassic Period** gave rise to birds which got modernised in the **Cretaceous Period**. The study of birds is known as **ornithology**. **Dr. Salim Ali** is a world famous **Indian ornithologist** ('Bird man of India').

Justify the statement - "Birds are glorified reptiles".

I. General Characters

- ❖ Body is streamlined and is distinguished into **head**, **neck**, **trunk** and **rudimentary tail**.

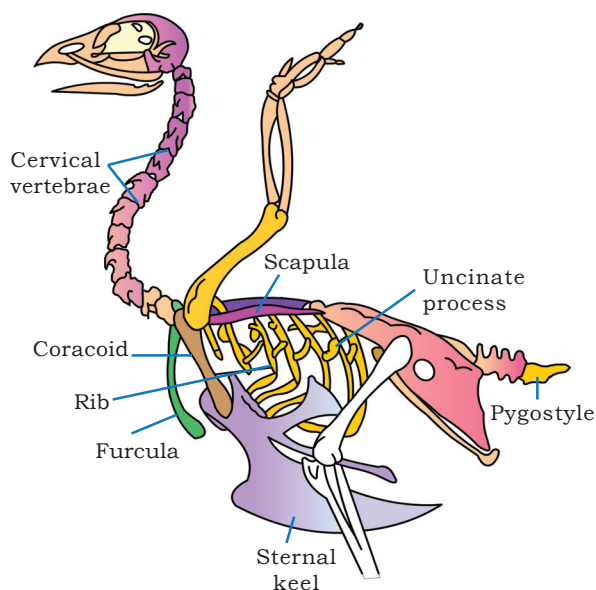


Fig. 4.24 Bird - skeleton

- ❖ The fore limbs are modified into 'wings' and hind limbs are adapted for walking, running, swimming, perching (sitting on a branch) etc.
- ❖ Skin is dry and devoid of glands, except for the oil gland or **preen gland** or **uropygial gland** at the base of the tail.
- ❖ Exoskeleton consists of epidermal feathers (a unique feature), scales on legs, claws on toes and horny covering on the beak (**rhamphotheca**).
- ❖ Endoskeleton is fully ossified. Long bones are hollow with air cavities (pneumatic). Skull is **monocondylic**. Vertebrae are heterocoelous. The last thoracic, lumbar, sacral and anterior few caudal vertebrae are fused to form a **synsacrum**. It is fused with

pelvic girdle to provide support to hind limbs. A few posterior most caudal vertebrae are fused to form the **pygostyle** that provides **support to the tail feathers**. Sternum has a **keel/carina** for the attachment of flight muscles (except in the ratite birds). Both the clavicles are fused with the interclavicle to form a 'V' – shaped bone, called **furcula** or '**wish bone**' or '**Merry thought bone**'. Ribs are **double headed** as in the crocodiles and therian mammals.

- ❖ All modern flying birds are provided with powerful **breast muscles** (flight muscles), chiefly the **pectoralis major** and **pectoralis minor**.
- ❖ **Teeth are absent** in the extant birds. Oesophagus is often dilated into a crop for the storage of food. Stomach is usually divided into glandular **proventriculus** and muscular **gizzard** (grinding mill). Cloaca is **three chambered** as in reptiles.
- ❖ Respiratory system consists of compact, spongy, undistensible lungs 'without alveoli'. The lungs are associated with **air sacs**. Air sacs facilitate 'continuous oxygenation' of blood and 'pneumaticity' of bones (a unique feature). Voice box is **syrinx** that lies at the junction of the trachea and bronchi.
- ❖ Heart is four chambered. **Sinus venosus** and **conus arteriosus** are absent as in mammals. Only **right systemic arch** is present. **Renal portal system** is reduced. Erythrocytes are nucleate as in reptiles.

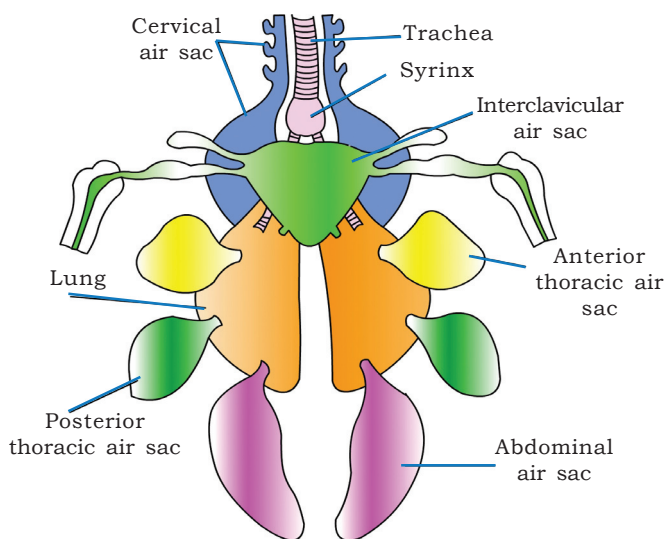


Fig. 4.25 Bird - air sacs

From your knowledge of the flying birds, what chief feature might have contributed for their successful aerial mode of life?

- ❖ Kidneys are **metanephric** and **three-lobed**. Urinary bladder is absent⁷ except in ostrich. Birds are **uricotelic** like the reptiles.
- ❖ Brain is large. Olfactory lobes are reduced. Cranial nerves are 12 pairs.
- ❖ Eyes possess **sclerotic plates**. A comb shaped vascular 'pecten' projects from the retina into the **vitreous humour** (except in kiwi). Middle ear has a single ear ossicle, the **columella auris**. Olfactory sense is usually poor, except in kiwi.
- ❖ Sexes are separate. Testes are paired, but the ovary and oviduct of the right side are almost completely atrophied in a **mature female**. Copulatory organ is absent in males except in **ratites, ducks, geese**, etc.

⁷ Loss of urinary bladder is believed to be one of the adaptations to reduce body weight of a bird.

- ❖ All birds are oviparous. Eggs are megalecithal and cleidoic. Fertilization is internal. Cleavage is **meroblastic** and **discoidal** as in reptiles. Hatchlings are **altricial** in the flying birds and **precocial** in the flightless birds.

e.g. *Corvus* (crow), *Columba* (pigeon), *Psittacula* (parrot), *Pavo* (peafowl; National bird of India), *Aptenodytes* (penguin), *Neophron* (vulture), *Coracias benghalensis* (blue jay; State bird of A.P.); *Struthio* (ostrich)



Struthio



Aptenodytes



Pavo



Psittacula



Coracias



Neophron

Fig. 4.26 Birds



Fig. 4.27 *Archaeopteryx* (imaginary diagram)

II. The 'Lizard bird'

Archaeopteryx lithographica (ancient bird; lizard bird) is the Jurassic fossil bird. It possesses some reptilian and avian characters, hence considered the '**connecting link**' between reptiles and birds.

Do you consider *Archaeopteryx* the 'connecting link' between the 'Belly creepers' and 'the Masters of air'. If so, how?

III. Ratitae / Palaeognathae

They are modern flightless running birds. They are 'discontinuous' in their distribution like the **lung fishes** and **marsupials**. They are characterized by the presence of reduced wings, a raft like sternum without keel and males with penis. They do not possess **syrinx**, **clavicles** and usually **preen gland**. They show discontinuous distribution.

e.g. *Struthio camelus* (African ostrich), *Kiwi* (National bird of New Zealand), *Rhea* (American ostrich), *Dromaeus* (Emu), *Casuarius*.

4.6.4 Class - Mammalia (L. *mamma*-breast)

The class mammalia consists of extinct and extant animals including humans. They are hairy, homeothermic amniotes. They nourish their young with milk secreted by specialised mammary glands from which the word 'mammalia' was coined. They show lot of variations in size and shape. The smallest of all the mammals is the recently discovered **ketti's hognosed bat** / '**Bumblebee bat**' in Thailand (weight- 2gms) and the biggest is **Balaenoptera musculus** (blue whale) weighing more than 130 tons. Extinct 'mammal like' **therapsid reptiles** gave rise to the early mammals in the **Triassic period**. **Coenozoic era** is referred to as '**the age of mammals**' as they became the predominant tetrapod group during this era. Mammals fall under three categories namely **monotremes** (egg laying mammals), **marsupials** (pouched mammals) and **eutherians** (true placentals). Intelligence and 'parental care' have reached their climax in mammals. The study of mammals is known as **mammology**.



Fig. 4.28 Therapsid reptile (imaginary diagram)

General characters

- ❖ The body is divided into head, neck, trunk and tail.
- ❖ Body is covered by hair, which is one of the unique characters of mammals. Hair is reduced in **whales**, **armadillos** etc. Hair is present at least in some stage of development in all mammals.
- ❖ Skin is glandular and consists of **sweat glands** (sudoriferous), **scent glands** and **sebaceous glands**. **Mammary glands** are modified sweat glands.
- ❖ Skull is **dicondylic**. Each half of the lower jaw consists of a 'single' bone, the **dentary**. Most mammals have **seven cervical vertebrae**, six in *Choloepus* (two-toed sloth) and *Trichechus* (manatee) and nine in *Bradypus* (three-toed sloth). Sacral vertebrae are **two to five**. Vertebrae are of the **amphiplatyan** type (centrum is flat at both faces). Ribs are double-headed.
- ❖ Buccal cavity is separated from the upper nasal cavity by a secondary palate. Teeth are **thecodont**, **heterodont** and **diphyodont**. Four pairs of salivary glands are present in association with the buccal cavity (3 pairs in man).
- ❖ Respiratory gaseous exchange occurs through lungs. Glottis is guarded by **epiglottis**. **Larynx** is the sound-producing organ.
- ❖ Heart is four-chambered. Oxygenated and de-oxygenated types of blood are completely separated. Only the left systemic arch is present. **Renal portal system** is absent. Mature RBC is circular, biconcave and **enucleate**.
- ❖ Mammals have a relatively large brain when compared to that of other animals in relation to body size. The four optic lobes constitute **corpora quadrigemina**. The two halves of cerebrum are connected by **corpus callosum**. The CNS is enveloped

by three meninges. The middle meninx called, **arachnoid membrane**, is present in mammals only. Cranial nerves are twelve pairs.

- ❖ Eyes have movable eye lids with 'eye lashes'. External ear has a large fleshy and cartilaginous flap called **pinna**. Middle ear possesses three ear ossicles. They are **malleus**, **incus** and **stapes**. **Cochlea** of the internal ear is **spirally coiled** and bears the '**organ of Corti**' which is the receptor of sound.
- ❖ Functional kidneys are bean shaped and **metanephric**. Nephron has **loop of Henle** which helps in the formation of concentrated urine. Mammals are 'ureotelic'.
- ❖ Sexes are separate. Sexual dimorphism is generally well-marked. Testes are usually found in scrotal sacs outside the abdomen. Intra-abdominal testes are found in the **monotremes**, **cetaceans** (whales and dolphins), **sea cows**, **elephants**, etc. Males have copulatory organ called **penis**.
- ❖ Fertilization is internal. Mammals, except the monotremates, are viviparous. Development is intra-uterine. Developing embryo is attached to the uterine wall of the mother by a **placenta**, for nutrition and respiration (absent in the monotremes).

Examples

Ornithorhynchus (duckbilled platypus), *Macropus* (Kangaroo), *Pteropus* (flying fox), *Camelus* (camel), *Macaca* (monkey), *Rattus* (rat), *Canis* (dog), *Felis* (cat), *Elephas* (elephant), *Equus* (horse), *Delphinus* (common dolphin), *Balaenoptera* (blue whale), *Panthera tigris* (tiger; National Animal of India), *Panthera leo* (lion), *Antelope* (state animal of A.P)

Do you know - The monotremes inspite of being 'egg laying mammals', also feed their young ones with milk?

Do you know - The gestation is very short in marsupials and the young one after birth crawls into the pouch and grows sucking milk. Some call it a 'mammary foetus'.



Ornithorhynchus



Macropus



Pteropus



Balaenoptera



Panthera tigris



Antelope

Fig. 4.29 Mammals

GLOSSARY

CHORDATA

Atrium: One of the chambers of the heart; the tympanic cavity of ear; the spacious ectoderm-lined cavity, enclosing the pharynx in most tunicates and cephalochordates.

Creatine phosphate: A high-energy phosphate compound, found in the muscles of vertebrates and some invertebrates, used to regenerate ATP.

Endostyle: Mucous secreting longitudinal ciliated groove present on the ventral wall of the pharynx of tunicates, cephalochordates, and larvae of the jawless fishes, useful for accumulating and moving food particles to the oesophagus.

Filter feeding: Any feeding process by which particulate food is filtered by the ciliary action from the water in which it is suspended e.g. Protochordates, Sponges and Bivalves

Mesonephros: It is the functional kidney of the adult fishes, amphibians and the embryos of the 'amniotes'.

Pelagic animals: Animals living in open oceans or seas, freely floating.

Portal system: A system of large veins beginning and ending in a plexus of capillaries e.g. hepatic and renal portal systems of vertebrates.

Retrogressive metamorphosis: Metamorphosis of a larva with advanced characters into a degenerate adult e.g. Ascidians, *Sacculina*.

Solenocyte: It is a tubular terminal flame cell like structure of a protonephridium. It has one or more flagella, which drive the excretory fluid along the tubule. They are found in cephalochordates.

PISCES

Acrodont: Having teeth attached to the edge of the jawbone without sockets.

Claspers: They are formed from the posterior portion of pelvic fins in male cartilaginous fish. They serve as intromittent organs used to channel semen into the female's cloaca during mating.

Coelacanth: An ancient, extant bony fish of the group Rhipidistia which was thought to be extinct until 1938, but was rediscovered in 1938, off the coast of South Africa. At present, there are two living species of the genus 'Latimeria'.

Ctenoid scale: A fish scale having marginal projections that resemble the teeth of a comb, found in many teleost fishes.

Cycloid scale: A fish scale which is thin and shows concentric lines of growth, without serrations on the margin, found in lung fishes and some teleost fishes.

Dipnoi: They are a group of fish commonly known as 'lung fishes', their lungs being modified air bladders. e.g. *Protopterus*, *Neoceratodus* and *Lepidosiren*.

Ganoid scales: Thick, bony scales of some primitive bony fishes such as *Acipenser*.

Ostracoderms: Extinct palaeozoic fishlike jawless vertebrates, characterized by a heavily armoured body; considered the ancestors of the jawed fishes.

Placoid scales: The type of scales found in cartilaginous fishes, with a basal plate of dentine embedded in the skin and a backward-pointing spine, tipped with vitrodentine.

Polyphyodont: A type of dentition in which the teeth are naturally shed many times

and replaced during the lifetime of vertebrates.

Ray finned fishes: They are a highly diverse group of aquatic vertebrates. Over half of all living vertebrate species are ray finned fishes.

AMPHIBIA

Alveolus: A small cavity or pit, such as a microscopic air sac of the lungs, terminal part of an alveolar gland, or bony socket of a tooth in the jaws of mammals and crocodiles.

Amplexus: The copulatory embrace of frogs or toads during which the male and female shed their gametes.

Chyme: Semifluid mass of partly digested food formed in the stomach.

Columella auris: A small rod like bone in the middle ear of frogs, reptiles and birds that transmits sound to inner ear; homologous to the mammalian stapes (modification of the hyomandibula of the fishes).

Conus arteriosus: It is a single, wide arterial vessel leaving the ventricle and passing ventrally over the right atrium. It is absent in the amniotes.

Dura mater: It is the outermost, toughest, and the most fibrous of the three membranes (meninges) covering the brain and the spinal cord.

Harderian gland: A gland associated with eye; the fluid it secretes varies in different groups of animals. In some animals it acts as an accessory gland to the lacrimal gland; secreting a fluid that lubricates the nictitating membrane.

Labyrinthodontia: A group of extinct amphibians typically resembles heavy-bodied salamanders and crocodiles. They have folded enamel and dentine at the base of conical teeth. e.g. Eryops.

Lacrimal glands: These are paired glands, one in each eye. They secrete watery fluid (tears), which contains lysozyme.

Opisthocoelous vertebra: In this type, the centrum is convex at the anterior face and concave at the posterior face; found in the urodeles.

Piamater: It is the delicate, inner most and vascularized membrane (meninx) enveloping the brain and spinal cord. It forms the choroid plexuses.

Sinus venosus: An enlarged region between the vena cavae and the right atrium. In the heart of frog, the pacemaker is the sinus venosus, (the 'SA node' in the heart of a mammal is believed to be an evolutionary remnant of the sinus venosus.)

REPTILIA

Allantois: It is one of the four extra embryonic membranes of the amniotes. It participates in respiration and excretion in sauropsids and also in the formation of placenta in most of the therians.

Amnion: It is the innermost of the extra embryonic membranes. It is a fluid-filled sac enclosing the embryo in the amniotes. It is protective in function (protection from shock and desiccation).

Chorion: It is the outermost extra embryonic membrane that surrounds the embryo of amniotes. It participates in the formation of placenta along with allantois in the placental mammals.

Hemipenis (plural: hemipenes): It is one of the pairs of intromittent organs of male squamates (snakes and lizards)

Meroblastic cleavage: It is a partial cleavage which occurs in the megalecithal eggs. The cleavages are

restricted to a disc like cytoplasmic area near the animal pole. e.g. sauropsids and monotremes.

Metanephros: It is the functional kidney of adult amniotes. It is the most efficient type of kidney.

Sauropsida: It is a group of amniotes, which includes all extinct and living reptiles and birds .

Temporal fossae: These are shallow depressions in the temporal regions of the skull of many reptiles, which provide space for the insertion of muscles.

AVES

Altricial hatchling: It is the hatchling of carinate bird which is 'incapable of moving around on its own soon after hatching'.

Carinate birds: These are the birds with a 'keeled sternum' for the insertion of flight muscles. e.g. flying birds.

Heterocoelous vertebra: It is the type of vertebra in which the articulating surfaces of the vertebral centrum are 'saddle-shaped'.

Pecten: It is a pigmented, vascular, and comblike process that projects into the vitreous humour from the retina at the point of entrance of the optic nerve in the eyes of birds and some reptiles.

Precocial young: It is the hatchling of a ratite bird, which is 'capable of moving around on its own soon after hatching'.

Theropods: The extinct bipedal carnivorous dinosaurs that gave rise to birds at the beginning of the Jurassic period.

Vitreous humour: It is the clear gel that fills the space between the lens and the retina of the eyeball of humans and other vertebrates

MAMMALIA

Cochlea: A tubular cavity of the inner ear containing the essential organs of hearing; occurs in crocodiles, birds and mammals. In the eutherians, it is spirally coiled with 'organ of Corti'- a specialized region for sound perception.

Diphyodont dentition: It is the type of dentition in which two sets of teeth are formed. They are deciduous and permanent sets of teeth which are formed successively.

Heterodont dentition: It is the type of dentition in which the teeth are differentiated into cutting, tearing, and grinding teeth.

Malleus: The outer ear ossicle which is attached to the tympanum in the middle ear of mammals (modification of the articular bone of the ancestors).

Placenta: The vascular structure, formed from embryonic and maternal tissues , through which the embryo and foetus are nourished while in the uterus.

Sebaceous glands: These are a type of mammalian epidermal glands associated with the hair follicles. They secrete 'sebum' which keeps the skin and hair smooth and lustrous.

Stapes: Stirrup-shaped, inner most ossicle of the middle ear in mammals (modification of hyomandibula); Smallest bone in the body of a mammal.

Sudoriferous glands: These are epidermal glands of the skin whose secretion (sweat) helps in excretion and thermoregulation; found only in mammals.

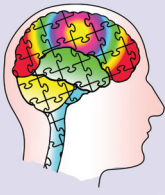
Therapsids: These are mammal-like extinct reptiles that gave rise to mammals during the 'Triassic period'.

VERY SHORT ANSWER QUESTIONS

- List out the characters shared by chordates and echinoderms.
- Write four salient features of cyclostomes.
- What is the importance of endostyle in lancelets and ascidians?
- Name the type of caudal fin and scales that are present in a shark and *Catla*, respectively.
- What is the importance of air bladder in fishes?
- How do you justify the statement-heart in fishes is a branchial heart.
- What are claspers? Which group of fishes possesses them?
- How does the heart of an amphibian differ from that of a reptile?
- Name the structures that appeared for the first time in amphibians, in the course of evolution.
- How do you distinguish a male frog from a female frog?
- What is force pump in frog? Why is it named so?
- What are corporabigemina? Mention their chief function.
- Distinguish between mesorchium and mesovarium.
- Distinguish between milt and spawn.
- What are the Golden ages of the first jawed vertebrates and the first amniotes?
- Name two poisonous and non-poisonous snakes found in south India.
- In which features does the skin of a reptile differ from that of a frog.
- Describe a cat and a lizard on the basis of their chief nitrogenous wastes excreted.
- Name the four extra embryonic membranes.
- What are Jacobson's organs? What is their function?
- What are pneumatic bones? How do they help birds?
- What is 'wish bone'? What are the skeletal components that form it?
- What is continuous oxygenation of the blood? How is it made possible in birds?
- Distinguish between the crop and the gizzard in birds.
- Distinguish between altricial and precocial hatchlings.
- In which group of animals do we find three ear ossicles on each side and what are their names from the inner most to the outermost?
- How does a mature RBC of a mammal differ from that of other vertebrates?
- Name the characteristic type of vertebrae found in reptiles, birds and mammals.
- Name the three meninges. In which group of animals do you find all of them?
- Name the vertebrate groups in which renal portal system is absent.

SHORT ANSWER QUESTIONS

1. Give three major differences between chordates and non-chordates, and draw the sketch of a chordate's body showing those features.
2. Name the four hallmarks of chordates, and explain the principal function of each of them.
3. Describe the features of a tunicate that reveals its chordate identity .
4. Compare and contrast sea squirts and lancelets.
5. List out eight characteristics that help distinguish a fish from the other vertebrates.
6. Compare and contrast cartilaginous and bony fishes.
7. Describe the structure of the heart of frog.
8. Write eight salient features of the class Amphibia.
9. Describe the male reproductive system of frog with the help of a labelled diagram.
10. Write short notes on organs of special senses in frog.
11. List out the salient features of exo and endoskeleton in reptiles.
12. List out the extant orders of the Class Reptilia. Give two examples for each Order.
13. What are the modifications that are observed in birds that help them in flight?
14. What are the features peculiar to ratite birds? Give two examples of ratite birds.
15. Mention the most important features of nervous system and sense organs in mammals.
16. Write short notes on the following features of the eutherians.
 - a) Dentition.
 - b) Endoskeleton.
17. Give an example for each of the following.
 - a) A viviparous fish
 - b) A fish possessing electric organs
 - c) A fish possessing poison sting
 - d) an organ which regulates buoyancy in the body of a fish
 - e) An oviparous animal with milk producing glands
18. Mention two similarities between
 - a) Aves and mammals
 - b) A frog and a crocodile
 - c) A lizard and a snake
19. Name the following animals
 - a) A limbless amphibian
 - b) The largest of all living animals
 - c) An animal possessing dry and cornified skin
 - d) 'National Animal' of India
20. Write the generic names of the following.
 - a) An oviparous mammal
 - b) Flying fox
 - c) Blue whale
 - d) Kangaroo.



For ignited minds

- NOT FOR EVALUATION

1. *Amphioxus* is described as a typical chordate. When the urochordates are also chordates (protochordates along with the cephalochordates), why can't we call the urochordates also 'typical' chordates?
2. There is a marked morphological difference between the 'jawless fishes' and 'jawed fishes', with reference to their movement in water. Can you guess what it could be?
3. Our heart receives deoxygenated blood through the caval veins. The heart of a fish too, receives deoxygenated blood. Why do scientists describe the heart of fish only, as a venous heart? Why not ours too?
4. Why did Romer describe the 'lung fishes' as the 'uncles of amphibians'? Can you throw light on the evolutionary considerations that led him state so?
5. In terms of expenditure of energy, which group of fishes will have to spend more energy to keep afloat at a particular level in the column of water? - the cartilaginous fishes or the bony fishes.
6. If a frog is kept on land for some time (by not allowing it to go into water), what do you think will happen? If somebody suggests-it dies, do you agree with him? Give reason.
7. The larva of frog has a long coiled gut. When it metamorphoses into a young frog what will happen to the gut? Give reason for your assumption.
8. When the egg of a frog develops into a tadpole larva, why should not be the same case, with respect to the egg of a reptile, bird or an egg laying mammal?
9. Amniotes are well adapted for terrestrial life. How can you support the popular belief that the ancestors of the amniotes were aquatic organisms? (Clue: Biogenetic law)
10. Amniotes possess mesonephric kidneys during their embryonic stage for some time. Later they are replaced by the metanephric kidneys. However certain remnants of the embryonic mesonephric system persist in the male amniotes. What are those remnants in your view?
11. Reptiles are uricotelic. What do you expect a marine turtle to be - ammonotelic?/ureotelic? uricotelic? Can you substantiate your answer with probable reasons?
12. You have read the units on Aves and Mammalia in that order in your text book. Do you agree with that order? If so, what is the basis? If not, how do you explain?
13. You know that nitrogenous waste products in animals include ammonia, urea, uric acid, guanine etc. How do you account for the reasons why some organisms send waste in the form of ammonia, others urea and yet others uric acid?
14. If an egg of a hen is coated with varnish (egg A) and another one non-coated normal egg (egg B) are incubated for 21 days, what do you expect regarding the outcome or end result.
15. Why do marsupials occur in Australia and America only? Why not in other places? What possible reasons can you think of? (Clue: Theory of continental drift)
16. Bats are not closely related to birds. Yet they can fly. What do you call such a phenomenon of evolution? (Clue: Expected answer is related to type of evolution)
17. When the R.B.C of all other vertebrates are nucleate, the highly evolved mammals possess enucleate R.B.C. (Nucleus is extruded during maturation). What physiological advantages do you see in such R.B.C. Give a logically sound answer.